

Multi-dimensional graph fractional Fourier transform and its application to data compression

Fang-Jia Yan, Bing-Zhao Li*

School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, 100081, China

ARTICLE INFO

Article history:

Available online 9 August 2022

Keywords:

Graph Fourier transform
Fractional Fourier transform
Graph signal processing
Multi-dimensional signal processing
Data compression

ABSTRACT

Many multi-dimensional signals appear in the real world, such as digital images and data that has spatial and temporal dimensions. Designing a transform method to process these multi-dimensional signals in graph fractional domain is a key challenge in graph signal processing. This paper investigates the novel transform for multi-dimensional graph signals defined on Cartesian product graph and studies several related properties. Our work includes: (i) proposing the two-dimensional graph fractional Fourier transforms using two basic graph signal processing methods i.e. based on Laplacian matrix and adjacency matrix; (ii) extending the two-dimensional transforms to multi-dimensional graph fractional Fourier transforms (MGFRFT). MGFRFT provides an additional fractional analysis tool for multi-dimensional graph signal processing; (iii) exploring the advantages of MGFRFT in terms of spectrum, directional characteristics and computational time; (iv) applying the proposed transform to data compression to highlight its utility and effectiveness.

© 2022 Elsevier Inc. All rights reserved.

1. Introduction

In many cases, data resides in irregular domains. Utilizing the underlying structure in the data is the core of many signal processing tasks [1–3]. To describe the structure and interpret the complex relationships of network datasets in non-Euclidean spaces, some fundamental signal processing methods have been extended to graph signals [2–8]. More specifically, graph signals are represented by the nodes of the graph, and the edges show the relationship between the function values on the nodes. Data from sensors [1], social networks [4], genetics [5], neuroscience [6] and image processing [9] can be effectively represented by graph signals. In recent years, graph signal analysis has gained considerable attention, giving rise to the graph-based transforms such as graph Fourier transform (GFT) [10–12], graph filters [13–15], sampling and recovery on graphs [16–18] and uncertainty principle [19–21]. Like classical signal processing, one of the main research topics of graph signal processing is to design transform methods to exploit the structure of the signal on the graph [10,12,22–32].

From the existing research results, GFT mainly includes two basic methods: (i) The first method is from Algebraic Signal Processing (ASP) theory. The adjacency matrix of weighted graph is introduced as shift operator to generate graph signal model. This

adjacency-based graph Fourier transform expands the graph signal based on the eigenvector of the adjacency matrix, and the eigenvalue of the adjacency matrix represents the corresponding frequency [10]. (ii) The second method is based on the graph Laplacian matrix, and it comes from the spectral graph theory. It uses the graph Laplacian matrix as the basic construction unit, studying the eigenvalues and eigenvectors of graph Laplacian matrix. This Laplacian-based graph Fourier transform is defined by these eigenvectors [22]. Similar to the classical Fourier transform, the GFT is inadequate for describing some applications or dealing with their underlying mathematical problems. As a result, some new transform methods, such as the spectral graph wavelet transform (SGWT) [24], the windowed graph Fourier transform (WGFT) [23], joint (time and graph) Fourier transform (JFT) [25], the fractional Fourier transform (FRFT) on graphs [27,29] and the windowed graph fractional Fourier transform (WGFRFT) [30,31] have been introduced to address the shortcoming of the GFT. Particularly, the multi-dimensional graph Fourier transform (MGFT) has been proposed to solve the problem that the GFT fails to show the correct spectrum of multi-dimensional (m-D) graph signals which defined on Cartesian product of graphs [26,33,34]. MGFT can rearrange the 1-D spectrum obtained by the GFT into the m-D frequency domain, and provide the m-D spectrum of the signal [26].

In order to extract local features of graph signals more effectively, fractional order is also introduced to graph signal processing [27–32,35,36]. The fractional order has gained considerable attention in the last 20 years or so because of its important applica-

* Corresponding author.

E-mail address: li_bingzhao@bit.edu.cn (B.-Z. Li).

tions in signal analysis, optics, and signal recovery [37–41]. This new parameter provides flexibility in processing data and enables performance improvements. The graph fractional domain is a combination of the graph spectrum domain and fractional transform domain, and there are few research results about transform methods [27,29–32]. The definition of graph fractional Fourier transform (GFRFT) based on graph adjacency matrix is presented in [27]. Furthermore, a new spectral graph fractional Fourier Transform (SGFRFT) is proposed in [29]. Based on SGFRFT, we introduced the windowed graph fractional Fourier transform and studied related fast algorithm [30,31]. Joint time-vertex fractional Fourier transform (JFRT) provides an additional fractional analysis tool for joint time-vertex processing by extending both temporal and vertex domain Fourier analysis to fractional orders [32]. These transforms all show advantages in revealing the local characteristics of the graph signal. However, GFRFT and SGFRFT have the same disadvantage as GFT in displaying the spectrum of multi-dimensional (m-D) graph signals, that is, they ignore the product structure in the graph.

If a graph has a large number of nodes, the signal processing operations defined on the graph require more computational complexity. For example, the complexity of the SGFRFT using an eigendecomposition is $O(N^3)$ for a graph with N nodes [29]. Whenever the underlying graph can be decomposed into two or more factor graphs with fewer nodes, the computational cost of these operations can be reduced significantly. The Cartesian product graph can represent multi-domain data effectively, and some signal processing method based on Cartesian product graph is proposed [26,33,34,42–45]. However, GFRFT and SGFRFT of product graphs does not have directional characteristics along with each factor graph. To illustrate directional characteristics of the multi-dimensional graph signal and reduce computational complexity in graph fractional domain, we propose a new transform. This paper is organized as follows. Section 2 overviews two basic methods of GFT and fractional Fourier transform on graphs. In Section 3, we define the two-dimensional graph fractional Fourier transforms based on Laplacian matrix and adjacency matrix respectively. Then we generalize the two-dimensional method to m-dimensional to obtain the multi-dimensional graph fractional Fourier transform (MGFRFT) in Section 4. In addition, we show the advantages of our new transform and provide its properties. Finally, we apply MGFRFT to data compression to clarify its application potentials in Section 5.

2. Preliminaries

In this section, preliminaries to GFTs and fractional Fourier transforms on graph are given. First of all, we define a graph. A finite set of vertices $\mathcal{V} = \{v_0, \dots, v_{N-1}\}$, where $N = |\mathcal{V}|$ is the number of nodes, a set of edges $\mathcal{E} = \{(i, j) | i, j \in \mathcal{V}, j \sim i\} \subseteq \mathcal{V} \times \mathcal{V}$, and a weighted adjacency matrix W form a weighted graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, W\}$.

2.1. Graph Fourier transform

2.1.1. Laplacian-based graph Fourier transform

We assume that the graph is undirected with positive edge weights. The non-normalized graph Laplacian is a symmetric difference operator $\mathcal{L} = D - W$ [22], where D is a diagonal degree matrix of $\mathcal{G}$. Let $\{\chi_0, \chi_1, \dots, \chi_{N-1}\}$ be the set of orthonormal eigenvectors. Suppose that the corresponding Laplacian eigenvalues are sorted as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1} := \lambda_{\max}$. Therefore

$$\mathcal{L} = \chi \Lambda \chi^H, \quad (1)$$

where

$$\chi = [\chi_0, \chi_1, \dots, \chi_{N-1}], \quad (2)$$

and the diagonal matrix is $\Lambda = \text{diag}([\lambda_0, \lambda_1, \dots, \lambda_{N-1}])$. The superscript H represents the Hermitian transpose operation.

The graph signal f is defined as binding a scalar value to each node through the function $f: \mathcal{V} \rightarrow \mathbb{R}$. Using the definition of Laplacian-based graph Fourier transform (L-GFT) as in [22], the L-GFT of f is

$$\hat{f}(\ell) = \langle f, \chi_\ell \rangle = \sum_{n=0}^{N-1} f(n) \chi_\ell^*(n), \ell = 0, 1, \dots, N-1, \quad (3)$$

where $*$ denotes complex conjugate. The inverse L-GFT is given by

$$f(n) = \sum_{\ell=0}^{N-1} \hat{f}(\ell) \chi_\ell(n), n = 0, 1, \dots, N-1. \quad (4)$$

2.1.2. Adjacency-based graph Fourier transform

The adjacency-based graph Fourier transform (A-GFT) is given by the eigenvectors of the adjacency matrix W [10]. The Jordan decomposition of W is

$$W = \mathbf{V} \mathbf{J}_W \mathbf{V}^{-1}, \quad (5)$$

the A-GFT of a graph signal $\mathbf{f}$ can be defined as [10]:

$$\hat{\mathbf{f}} = \mathbf{V}^{-1} \mathbf{f}, \quad (6)$$

where $\mathbf{V}^{-1}$ is the A-GFT matrix.

The inverse A-GFT is defined as

$$\mathbf{f} = \mathbf{V} \hat{\mathbf{f}}. \quad (7)$$

Remark 1. In order to distinguish A-GFT from L-GFT, we use bold $\mathbf{f}$ to represent signals in adjacency-based method.

2.2. Fractional Fourier transform on graphs

2.2.1. Spectral graph fractional Fourier transform

Similar to L-GFT, the SGFRFT is also related to the graph Laplacian operator. The graph fractional Laplacian operator $\mathcal{L}_\alpha$ is defined by $\mathcal{L}_\alpha = \kappa R \kappa^H$, where α is the fractional order, $0 < \alpha \leq 1$ [29]. Note that

$$\kappa = [\kappa_0, \kappa_1, \dots, \kappa_{N-1}] = \chi^\alpha, \quad (8)$$

and

$$R = \text{diag}([r_0, r_1, \dots, r_{N-1}]) = \Lambda^\alpha, \quad (9)$$

so that

$$r_\ell = \lambda_\ell^\alpha, \ell = 0, 1, \dots, N-1. \quad (10)$$

In the follow-up part of this paper, computing the α power of a matrix always uses matrix power function.

The SGFRFT of signal f building on the graph $\mathcal{G}$ is defined by [29]:

$$\hat{f}_\alpha(\ell) = \langle f, \kappa_\ell \rangle = \sum_{n=0}^{N-1} f(n) \kappa_\ell^*(n), \ell = 0, 1, \dots, N-1, \quad (11)$$

when $\alpha = 1$, the SGFRFT degenerates into standard L-GFT.

The inverse SGFRFT is given by

$$f(n) = \sum_{\ell=0}^{N-1} \hat{f}_\alpha(\ell) \kappa_\ell(n), n = 0, 1, \dots, N-1. \quad (12)$$

2.2.2. Graph fractional Fourier transform

Just like A-GFT, the GFRFT comes from adjacency matrix. Further, the A-GFT matrix $\mathbf{V}^{-1}$ is decomposed as [27]:

$$\mathbf{V}^{-1} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1},$$

and the definition of the GFRFT matrix is [27]

$$(\mathbf{V}^{-1})^\alpha = \mathbf{P}\mathbf{J}^\alpha\mathbf{P}^{-1}.$$

The GFRFT of graph signal $\mathbf{f}$ can be written in the matrix form [27]:

$$\hat{\mathbf{f}}_\alpha = (\mathbf{V}^{-1})^\alpha \mathbf{f}, \quad (13)$$

where α is the fractional order.

The inverse GFRFT is defined as:

$$\mathbf{f} = \mathbf{V}^\alpha \hat{\mathbf{f}}_\alpha. \quad (14)$$

2.3. Cartesian product graph

Cartesian product graph is a type of graph multiplication [42]. Consider two graphs $\mathcal{G}_1 = \{\mathcal{V}_1, \mathcal{E}_1, W_1\}$ and $\mathcal{G}_2 = \{\mathcal{V}_2, \mathcal{E}_2, W_2\}$. $\mathcal{L}_1$ and $\mathcal{L}_2$ are the Laplacian matrix of $\mathcal{G}_1, \mathcal{G}_2$. First, $\square$ represents the Cartesian product between graphs. $\mathcal{G}_1 \square \mathcal{G}_2$ is the Cartesian product graph with vertex set $\mathcal{V}_1 \times \mathcal{V}_2$ where $\mathcal{V}_1 = \{0, \dots, N_1 - 1\}$, $\mathcal{V}_2 = \{0, \dots, N_2 - 1\}$. Then, the edge set of $\mathcal{G}_1 \square \mathcal{G}_2$ satisfying:

$$[\{i_1, j_1\} \in \mathcal{E}_1, i_2 = j_2] \quad \text{or} \quad [i_1 = j_1, \{i_2, j_2\} \in \mathcal{E}_2],$$

and the weight function w is

$$w((i_1, i_2), (j_1, j_2)) = W_1(i_1, j_1)\delta(i_2, j_2) + \delta(i_1, j_1)W_2(i_2, j_2).$$

δ is the delta function, i.e., $\delta(i, j)$ is 1 if $i = j$ and 0 otherwise. Based on this definition, it is easy to prove that Cartesian product graphs satisfy the following properties:

- (i) associative: as the graphs $(\mathcal{F} \square \mathcal{G}) \square \mathcal{H}$ and $\mathcal{F} \square (\mathcal{G} \square \mathcal{H})$ are naturally isomorphic.
- (ii) commutative: the graphs $\mathcal{G} \square \mathcal{H}$ and $\mathcal{H} \square \mathcal{G}$ are naturally isomorphic.

The graphs $\mathcal{G}_1, \mathcal{G}_2$ are factor graphs of $\mathcal{G}_1 \square \mathcal{G}_2$. The adjacency matrix $W_1 \oplus W_2$ and Laplacian matrix $\mathcal{L}_1 \oplus \mathcal{L}_2$ of $\mathcal{G}_1 \square \mathcal{G}_2$ can be expressed by those of its factor graphs.

$$W_1 \oplus W_2 = W_1 \otimes I_{N_2} + I_{N_1} \otimes W_2, \quad (15)$$

$$\mathcal{L}_1 \oplus \mathcal{L}_2 = I_{N_2} \otimes \mathcal{L}_1 + \mathcal{L}_2 \otimes I_{N_1}. \quad (16)$$

Here, the operator $\oplus$ represents the Kronecker sum and $\otimes$ represents the Kronecker product [46]. $\oplus$ is defined as $A \oplus B = A \otimes I_n + I_m \otimes B$ for matrices $A \in \mathbb{R}^{m \times m}$ and $B \in \mathbb{R}^{n \times n}$, where I_n is the identity matrix of size n . $\otimes$ is an operation on two matrices of arbitrary size resulting in a block matrix. If A is an $m \times n$ matrix and B is a $p \times q$ matrix, then the Kronecker product $A \otimes B$ is the $pm \times qn$ block matrix:

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}. \quad (17)$$

Then the eigenvalues of $\mathcal{G}_1 \square \mathcal{G}_2$ are all possible sums of $\lambda_i(\mathcal{G}_1) + \lambda_j(\mathcal{G}_2)$, with $0 \leq i \leq N_1 - 1$ and $0 \leq j \leq N_2 - 1$ [47].

We give an example of Cartesian product graph using in graph signal processing. In Fig. 1, the sensor network measurements are decomposed as the Cartesian product of sensor network and time series.

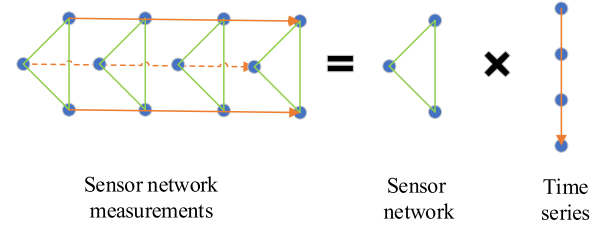


Fig. 1. Measurements of a sensor network.

3. Two-dimensional graph fractional Fourier transform

The eigenvalues of the graph Laplacian and the adjacency matrix act as the graph spectrum. In particular, when the Laplacian has non-distinct eigenvalues, the spectral functions are not well-defined, i.e., multi-valued [26]. There are similar problems in graph fractional domain. In order to show the spectrum more effectively, we propose the definition of two-dimensional graph fractional Fourier transform (2-D GFRFT) based on Laplacian matrix and adjacency matrix in this section. Though the spectrum obtained by these two 2-D GFRFTs is usually different for the same graph signal, the process of signal analysis is similar.

3.1. Laplacian-based two-dimensional graph fractional Fourier transform

Considering a product graph $\mathcal{G}_1 \square \mathcal{G}_2$ in Section 2.3, their Laplacian operators are defined as: $\mathcal{L}_1 = \chi_1 \mathbf{A}_1 \chi_1^H$, and $\mathcal{L}_2 = \chi_2 \mathbf{A}_2 \chi_2^H$.

The corresponding graph fractional Laplacian operators of $\mathcal{L}_1$ and $\mathcal{L}_2$ are:

$$\mathcal{L}_\alpha^{(1)} = \kappa^{(1)} R^{(1)} (\kappa^{(1)})^H,$$

where

$$\kappa^{(1)} = [\kappa_0^{(1)}, \kappa_1^{(1)}, \dots, \kappa_{N_1-1}^{(1)}] = \chi_1^\alpha,$$

and

$$R^{(1)} = \text{diag}([r_0^{(1)}, r_1^{(1)}, \dots, r_{N_1-1}^{(1)}]) = \mathbf{A}_1^\alpha.$$

And

$$\mathcal{L}_\alpha^{(2)} = \kappa^{(2)} R^{(2)} (\kappa^{(2)})^H,$$

where

$$\kappa^{(2)} = [\kappa_0^{(2)}, \kappa_1^{(2)}, \dots, \kappa_{N_2-1}^{(2)}] = \chi_2^\alpha,$$

and

$$R^{(2)} = \text{diag}([r_0^{(2)}, r_1^{(2)}, \dots, r_{N_2-1}^{(2)}]) = \mathbf{A}_2^\alpha.$$

Therefore, for $\mathcal{G}_1 \square \mathcal{G}_2$, we have the following eigendecomposition of $\mathcal{L}_\alpha^{(1)} \oplus \mathcal{L}_\alpha^{(2)}$ in graph fractional domain.

$$\begin{aligned} & (\mathcal{L}_\alpha^{(1)} \oplus \mathcal{L}_\alpha^{(2)}) \begin{pmatrix} \kappa_{\ell_1}^{(1)}(0)\kappa_{\ell_2}^{(2)}(0) \\ \kappa_{\ell_1}^{(1)}(0)\kappa_{\ell_2}^{(2)}(1) \\ \vdots \\ \kappa_{\ell_1}^{(1)}(N_1-1)\kappa_{\ell_2}^{(2)}(N_2-1) \end{pmatrix} \\ &= (r_{\ell_1}^{(1)} + r_{\ell_2}^{(2)}) \begin{pmatrix} \kappa_{\ell_1}^{(1)}(0)\kappa_{\ell_2}^{(2)}(0) \\ \kappa_{\ell_1}^{(1)}(0)\kappa_{\ell_2}^{(2)}(1) \\ \vdots \\ \kappa_{\ell_1}^{(1)}(N_1-1)\kappa_{\ell_2}^{(2)}(N_2-1) \end{pmatrix}. \end{aligned} \quad (18)$$

Define a signal on a Cartesian product graph as a “two-dimensional signal” [33]. Using the eigenvalues and eigenfunctions of $\mathcal{L}_\alpha^{(1)} \oplus \mathcal{L}_\alpha^{(2)}$ in (18), we propose the Laplacian-based two-dimensional graph fractional Fourier transform (L-2-D GFRFT).

Remark 2. The “two-dimensional signal” is not actual two-dimensional data. It is a two-dimensional graph signal, which is distributed on two-dimensional graph nodes. And this concept can be naturally generalized to multi-dimensional graph signal.

Definition 1. (Laplacian-based two-dimensional graph fractional Fourier transform) The L-2-D GFRFT of a two-dimensional signal $f: \mathcal{V}_1 \times \mathcal{V}_2 \rightarrow \mathbb{R}$ on a Cartesian product graph $\mathcal{G}_1 \square \mathcal{G}_2$ is

$$\hat{f}_\alpha(\ell_1 + \ell_2) = \sum_{n_1=1}^{N_1} \sum_{n_2=1}^{N_2} f(n_1, n_2) (\kappa_{\ell_1}^{(1)}(n_1))^* (\kappa_{\ell_2}^{(2)}(n_2))^*, \quad (19)$$

for $\ell_1 = 0, 1, \dots, N_1 - 1, \ell_2 = 0, 1, \dots, N_2 - 1$. And its inverse (IL-2-D GFRFT) is given by

$$f(n_1, n_2) = \sum_{\ell_1=1}^{N_1} \sum_{\ell_2=1}^{N_2} \hat{f}_\alpha(\ell_1 + \ell_2) \kappa_{\ell_1}^{(1)}(n_1) \kappa_{\ell_2}^{(2)}(n_2), \quad (20)$$

for $n_1 = 0, 1, \dots, N_1 - 1, n_2 = 0, 1, \dots, N_2 - 1$.

3.2. Adjacency-based two-dimensional graph fractional Fourier transform

Considering a product graph $\mathcal{G}_1 \square \mathcal{G}_2$, their adjacency matrices are defined as W_1, W_2 , and the eigendecomposition (5) of W_1 and W_2 is respectively $W_i = \mathbf{V}_i \mathbf{J}_{W_i} \mathbf{V}_i^{-1}, i \in 1, 2$.

The eigendecomposition of adjacency matrix $W_1 \oplus W_2$ in (15) is

$$W_1 \oplus W_2 = (\mathbf{V}_1 \otimes \mathbf{V}_2) (\mathbf{J}_{W_1} \otimes \mathbf{I}_{N_2} + \mathbf{I}_{N_1} \otimes \mathbf{J}_{W_2}) (\mathbf{V}_1 \otimes \mathbf{V}_2)^{-1}. \quad (21)$$

Using the above eigendecomposition, we define adjacency-based two-dimensional graph fractional Fourier transform (A-2-D GFRFT).

Definition 2. (Adjacency-based two-dimensional graph fractional Fourier transform) The A-2-D GFRFT of a two-dimensional signal $\mathbf{f}: \mathcal{V}_1 \times \mathcal{V}_2 \rightarrow \mathbb{R}$ on a Cartesian product graph $\mathcal{G}_1 \square \mathcal{G}_2$ is

$$\hat{\mathbf{f}}_\alpha = ((\mathbf{V}_1 \otimes \mathbf{V}_2)^{-1})^\alpha \mathbf{f} = (\mathbf{V}_1^{-1} \otimes \mathbf{V}_2^{-1})^\alpha \mathbf{f}. \quad (22)$$

Its inverse (IA-2-D GFRFT) is given by

$$\mathbf{f} = (\mathbf{V}_1 \otimes \mathbf{V}_2)^\alpha \hat{\mathbf{f}}_\alpha. \quad (23)$$

4. Multi-dimensional graph fractional Fourier transform

In this section, the multi-dimensional graph fractional Fourier transform (MGFRFT) based on Laplacian matrix and adjacency matrix is inductively defined by 2-D GFRFTs.

4.1. Laplacian-based multi-dimensional graph fractional Fourier transform

Given a natural number m , consider m graphs $\mathcal{G}_i = \{\mathcal{V}_i, \mathcal{E}_i, W_i\}$, $i = 1, 2, \dots, m$ with vertex set $\mathcal{V}_i$ and $\mathcal{V}_i = \{0, \dots, N_i - 1\}$, we have a Cartesian product graph $\mathcal{G}_1 \square \dots \square \mathcal{G}_m$. $\mathcal{L}_i$ is the Laplacian matrix of $\mathcal{G}_i$ and its eigendecomposition is $\mathcal{L}_i = \chi_i \Lambda_i \chi_i^H$. First, as mentioned in the Section 3.1, the graph fractional Laplacian operator of $\mathcal{L}_i$ is

$$\mathcal{L}_\alpha^{(i)} = \kappa^{(i)} R^{(i)} (\kappa^{(i)})^H,$$

where

$$\kappa^{(i)} = [\kappa_0^{(i)}, \kappa_1^{(i)}, \dots, \kappa_{N_i-1}^{(i)}] = \chi_i^\alpha,$$

and

$$R^{(i)} = \text{diag}([r_0^{(i)}, r_1^{(i)}, \dots, r_{N_i-1}^{(i)}]) = \Lambda_i^\alpha.$$

Because the Cartesian product graph $\mathcal{G}_1 \square \dots \square \mathcal{G}_m = ((\dots (\mathcal{G}_1 \square \mathcal{G}_2) \dots) \square \mathcal{G}_m)$, its Laplacian matrix $\mathcal{L}_\alpha^{(1)} \oplus \dots \oplus \mathcal{L}_\alpha^{(m)}$ satisfies:

$$\begin{aligned} & (\mathcal{L}_\alpha^{(1)} \oplus \dots \oplus \mathcal{L}_\alpha^{(m)}) \begin{pmatrix} \kappa_{\ell_1}^{(1)}(0) \kappa_{\ell_2}^{(2)}(0) \dots \kappa_{\ell_m}^{(m)}(0) \\ \kappa_{\ell_1}^{(1)}(0) \kappa_{\ell_2}^{(2)}(0) \dots \kappa_{\ell_m}^{(m)}(1) \\ \vdots \\ \kappa_{\ell_1}^{(1)}(N_1 - 1) \kappa_{\ell_2}^{(2)}(N_2 - 1) \dots \kappa_{\ell_m}^{(m)}(N_m - 1) \end{pmatrix} \\ &= (r_{\ell_1}^{(1)} + \dots + r_{\ell_m}^{(m)}) \\ & \times \begin{pmatrix} \kappa_{\ell_1}^{(1)}(0) \kappa_{\ell_2}^{(2)}(0) \dots \kappa_{\ell_m}^{(m)}(0) \\ \kappa_{\ell_1}^{(1)}(0) \kappa_{\ell_2}^{(2)}(0) \dots \kappa_{\ell_m}^{(m)}(1) \\ \vdots \\ \kappa_{\ell_1}^{(1)}(N_1 - 1) \kappa_{\ell_2}^{(2)}(N_2 - 1) \dots \kappa_{\ell_m}^{(m)}(N_m - 1) \end{pmatrix}. \end{aligned} \quad (24)$$

Definition 3. (Laplacian-based multi-dimensional graph fractional Fourier transform) The L-MGFRFT of a m -dimensional signal f on a Cartesian product graph $\mathcal{G}_1 \square \dots \square \mathcal{G}_m$ is

$$\begin{aligned} & \hat{f}_\alpha(\ell_1 + \dots + \ell_m) \\ &= \sum_{n_1=1}^{N_1} \sum_{n_2=1}^{N_2} \dots \sum_{n_m=1}^{N_m} f(n_1, \dots, n_m) (\kappa_{\ell_1}^{(1)}(n_1))^* \dots (\kappa_{\ell_m}^{(m)}(n_m))^*, \end{aligned} \quad (25)$$

for $\ell_i = 0, 1, \dots, N_i - 1$. And its inverse (IL-MGFRFT) is given by

$$\begin{aligned} & f(n_1, \dots, n_m) \\ &= \sum_{\ell_1=1}^{N_1} \sum_{\ell_2=1}^{N_2} \dots \sum_{\ell_m=1}^{N_m} \hat{f}_\alpha(\ell_1 + \dots + \ell_m) \kappa_{\ell_1}^{(1)}(n_1) \dots \kappa_{\ell_m}^{(m)}(n_m), \end{aligned} \quad (26)$$

for $n_i = 0, 1, \dots, N_i - 1$.

4.2. Adjacency-based multi-dimensional graph fractional Fourier transform

Using the same Cartesian product graph as in Section 4.1, the eigendecomposition of the adjacency operator W_i of graph $\mathcal{G}_i$ is $W_i = \mathbf{V}_i \mathbf{J}_{W_i} \mathbf{V}_i^{-1}$. Like (21), the eigendecomposition of adjacency matrix $W_1 \oplus \dots \oplus W_m$ is

$$\begin{aligned} & W_1 \oplus \dots \oplus W_m \\ &= (\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m) \left(\sum_{j=1}^m \mathbf{I}_{N_1 \dots N_{j-1}} \otimes \mathbf{J}_{W_j} \otimes \mathbf{I}_{N_{j+1} \dots N_m} \right) \\ & \times (\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1}. \end{aligned} \quad (27)$$

Using the above eigendecomposition, adjacency-based multi-dimensional graph fractional Fourier transform (A-MGFRFT) is defined.

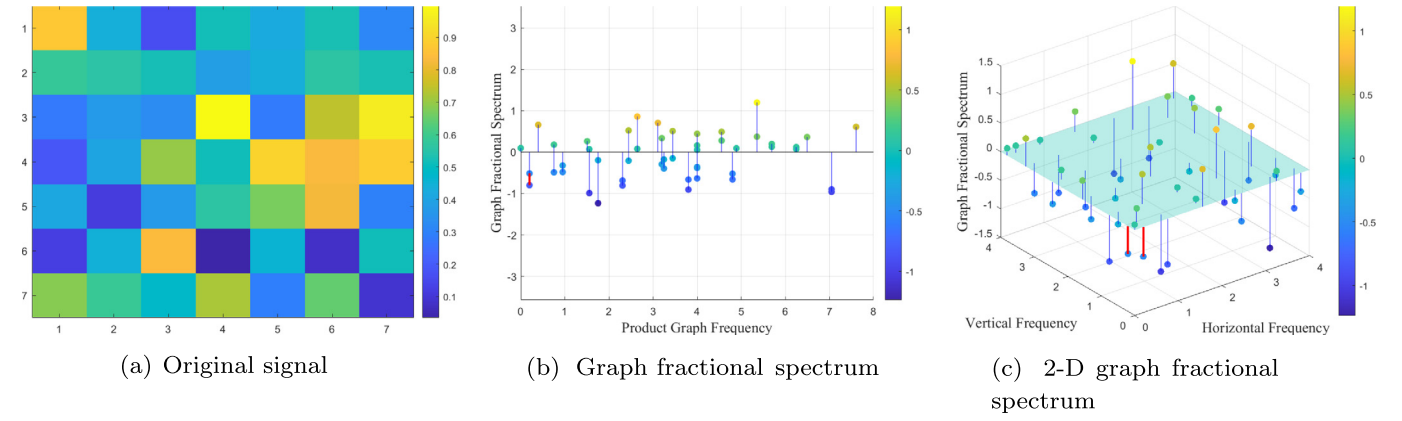


Fig. 2. 2-D graph signal and its spectrum. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Definition 4. (Adjacency-based multi-dimensional graph fractional Fourier transform) The A-MGFRFT of a m -dimensional signal $\mathbf{f}$ on a Cartesian product graph $\mathcal{G}_1 \square \dots \square \mathcal{G}_m$ is

$$\hat{\mathbf{f}}_\alpha = ((\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1})^\alpha \mathbf{f} = (\mathbf{V}_1^{-1} \otimes \dots \otimes \mathbf{V}_m^{-1})^\alpha \mathbf{f}. \quad (28)$$

Its inverse (IA-MGFRFT) is given by

$$\mathbf{f} = (\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^\alpha \hat{\mathbf{f}}_\alpha. \quad (29)$$

In graph signal processing, two classical forms of graph fractional Fourier transform are based on Laplacian matrix and adjacency matrix. For later use, we extend both classical definitions to the multidimensional case. According to different application requirements, we can choose the appropriate method. For example, if we know the adjacency matrix of graph structure, then we can use A-MGFRFT. If the graph is undirected, it is more convenient to use L-MGFRFT.

4.3. Essential properties of MGFRFT

Take the definition of A-MGFRFT as an example, some properties of MGFRFT are derived.

Property 1. When $\alpha = 0$, the A-MGFRFT reduces to identity transform. This property coincides with FRFT and GFRFT.

Property 2. When $\alpha = 1$, the A-MGFRFT is consistent with MGFT in [26]. That means that A-MGFRFT is the natural form of MGFT extending to the fractional order.

Property 3. Index additivity. Let $\mathbf{F}^\alpha = (\mathbf{V}_1^{-1} \otimes \dots \otimes \mathbf{V}_m^{-1})^\alpha$ be the A-MGFRFT matrix. Then it is index additive.

$$\mathbf{F}^\beta (\mathbf{F}^\alpha) = (\mathbf{V}_1^{-1} \otimes \dots \otimes \mathbf{V}_m^{-1})^{\beta+\alpha} = \mathbf{F}^{\beta+\alpha}. \quad (30)$$

Proof.

$$\begin{aligned} \mathbf{F}^\beta (\mathbf{F}^\alpha) &= ((\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1})^\beta ((\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1})^\alpha \\ &= ((\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1})^{\beta+\alpha} = (\mathbf{V}_1^{-1} \otimes \dots \otimes \mathbf{V}_m^{-1})^{\beta+\alpha} \\ &= \mathbf{F}^{\beta+\alpha}, \end{aligned} \quad (31)$$

where the second equality induces from the index additivity property of GFRFT [27]. $\square$

Property 4. Commutativity. That means the A-MGFRFT matrix satisfies:

$$\mathbf{F}^\beta (\mathbf{F}^\alpha) = \mathbf{F}^\alpha (\mathbf{F}^\beta). \quad (32)$$

Proof. By **Property 3**, we get:

$$\begin{aligned} \mathbf{F}^\beta (\mathbf{F}^\alpha) &= ((\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1})^\beta ((\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1})^\alpha \\ &= ((\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1})^\alpha ((\mathbf{V}_1 \otimes \dots \otimes \mathbf{V}_m)^{-1})^\beta = \mathbf{F}^\alpha (\mathbf{F}^\beta). \quad \square \end{aligned} \quad (33)$$

4.4. Advantage analysis of MGFRFT

For a signal on the Cartesian product graph, our MGFRFTs provide the following advantage over the traditional GFRFT and SGFRFT. We take the L-2-D GFRFT as an example to illustrate how the advantage is reflected in the graph signal processing. Since the spectrum of the multi-dimensional graph signal obtained by 1-D transforms is often multi-valued at a specific frequency and the GFRFT and SGFRFT can only show 1-d spectrum. These two traditional methods are not suitable for multi-dimensional graph signals. In Fig. 2, we visually display this shortcoming of 1-D transforms and the advantage of L-2-D GFRFT. Fig. 2(a) is a random 2-D graph signal on the grid graph which is a product of two path graphs. The x and y axis represent seven vertices of path graphs. The color of each node represents the intensity of the random signal. In Fig. 2(b), we see that at certain frequencies, the spectrum is double-valued. We use red line to give an example of double-valued spectrum. However, in Fig. 2(c), all the spectra are single-valued, and we use the same red color to show the double-valued spectrum in Fig. 2(b) can be identified and separated by our new transform. Therefore, the L-2-D GFRFT can solve the multi-valuedness of the spectral functions.

MGFRFT also enables us to analyze graph signals according to the frequency characteristics of each factor graph. In contrast, GFRFT and SGFRFT of product graphs cannot give such analysis. We illustrate this advantage in a synthetic example in Fig. 3. In Fig. 3(a), a random signal is placed on a 8×8 grid graph where it fluctuates drastically along the vertical path graph. As a result, we see that in Fig. 3(b) the spectrum at large vertical frequency is dominant. In comparison, another random signal is set to change gradually along the vertical path graph as in Fig. 3(c), and the spectrum at small vertical frequency is dominant as in Fig. 3(d).

Another advantage that can not be ignored is time consumption. As Cartesian product graph can represent multi-domain data effectively [42], when we use two or more factor graphs with fewer nodes to replace the original graph, the computational cost can be reduced significantly. Considering a Cartesian product graph $\mathcal{G}_1 \square \mathcal{G}_2$, the L-2-D GFRFT costs $O(N_1^3 + N_2^3)$ time, whereas the traditional SGFRFT costs $O(N_1^3 N_2^3)$ on the same graph. When we use factor graphs $\mathcal{G}_1$ and $\mathcal{G}_2$ instead of the original graph to calculate the eigendecomposition of the graph fractional Laplacian, it also takes less time.

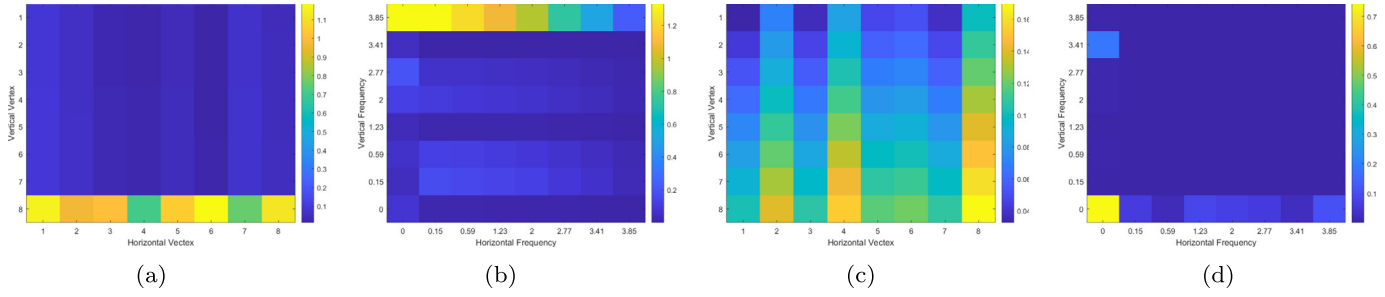


Fig. 3. Illustration of directional characteristics of MGFRFT. Fig. (a) represents a random signal on the 8×8 grid graph which drastically changes along the vertical path graph. Fig. (b) is the graph fractional spectrum of (a). Similarly, Fig. (c) represents a random signal on the 8×8 grid graph which changes gradually along the vertical path graph and Fig. (d) is its graph fractional spectrum.

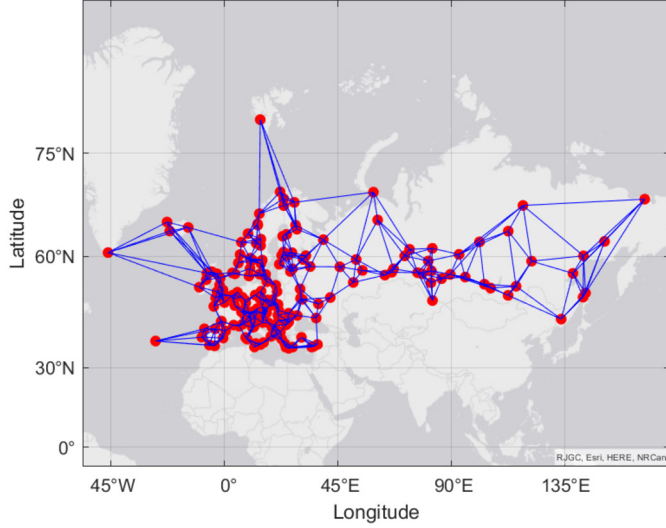


Fig. 4. The k NN graph $\mathcal{G}$ for weather observation station networks.

Compared with SGFRFT and GFRFT, MGFRFT provides additional flexibility through the fractional parameter α without increasing computational costs. And it can lead to some performance improvements in graph signal processing. In the next section we provide numerical examples of data compression where MGFRFT brings us a better performance.

5. Applications

To illustrate the effectiveness of the framework in MGFRFT, we consider an application to data compression. We use public real-world data from National Oceanic and Atmospheric Administration (NOAA)'s National Centers for Environmental Information (NCEI) where we randomly select 200 weather observation stations and pick out temperature records in 2020. The data is available in <https://www.ncei.noaa.gov/data/global-summary-of-the-day/access/2020/>.

The temperature records naturally form a path graph with 365 nodes, denoted by $\mathcal{T}$. To construct a graph model for stations, we adopt k NN method as shown in [48]. For each pair of stations, we compute the spherical distance via latitudes and longitudes. Specifically, let (θ_i, ϕ_i) and (θ_j, ϕ_j) be the coordinates for two stations i and j (in radian measure). The spherical distance is defined by

$$d_{ij} = \arccos \left(\cos(\theta_i) \cos(\theta_j) \cos(\phi_i - \phi_j) + \sin(\theta_i) \sin(\theta_j) \right). \quad (34)$$

Note that we drop the earth radius $R = 6,357$ km in the computation of spherical distance since it will cause large precision error when we use the Gaussian kernel.

Then we search the k -nearest-neighbors for each station based on the spherical distance. The k NN graph $\mathcal{G}$ is constructed by setting each node as a station and a pair of nodes is connected iff they are neighbors. Fig. 4 shows the k NN graph where k is set to be 5.

The weighted matrix W is defined by substituting the spherical distances into the Gaussian kernel. Let i and j be two nodes joined by an edge. The weight W_{ij} is set to be

$$W_{ij} = \frac{\exp(-\frac{d_{ij}^2}{\sigma^2})}{\sqrt{\sum_{k \sim i} \exp(-\frac{d_{ik}^2}{\sigma^2})} \sqrt{\sum_{k \sim j} \exp(-\frac{d_{jk}^2}{\sigma^2})}}, \quad (35)$$

where $\sim$ denotes that two nodes are incident, and σ^2 is a preset parameter.

The temperature data thus can be regarded as a signal f on the product graph $\mathcal{G} \square \mathcal{T}$ where each element $f(i, t)$ denotes the temperature at station i on day t . The compression process is performed as follows. We compute the MGFRFT of f using equation (25). Then we sort the coefficients according to their absolute values. Let $0 < \gamma < 1$ be the compression ratio. We retain γ fraction of largest coefficients and set remaining coefficients to be zero. Finally we perform the IMGFRFT on the resulting coefficients. Fig. 5 presents a 10×10 block of the original temperature data and compressed temperature data.

Four quantities are used to measure the compression loss. Let f_{com} be the compressed signal. The relative error (RE) is defined by

$$RE = \frac{\sum_{i \in \mathcal{G}, t \in \mathcal{T}} |f(i, t) - f_{com}(i, t)|}{\sum_{i \in \mathcal{G}, t \in \mathcal{T}} |f(i, t)|}. \quad (36)$$

The peak signal-to-noise ratio (PSNR) is defined by

$$PSNR = 10 \log_{10} \left(\frac{\max_{i \in \mathcal{G}, t \in \mathcal{T}} |f_{com}(i, t)|^2}{\frac{1}{|\mathcal{G}| |\mathcal{T}|} \sum_{i \in \mathcal{G}, t \in \mathcal{T}} |f(i, t) - f_{com}(i, t)|^2} \right). \quad (37)$$

Let $\bar{f} = \frac{1}{|\mathcal{G}| |\mathcal{T}|} \sum_{i \in \mathcal{G}, t \in \mathcal{T}} f(i, t)$ be the mean of the original signal and define $\bar{f}_{com}$ in the similar way for the compressed signal. The normalized root mean square (NRMS) is defined by

$$NRMS = \frac{\sqrt{\sum_{i \in \mathcal{G}, t \in \mathcal{T}} (f(i, t) - f_{com}(i, t))^2}}{\sqrt{\sum_{i \in \mathcal{G}, t \in \mathcal{T}} (f(i, t) - \bar{f})^2}}. \quad (38)$$

The correlation coefficient (CC) is defined by

$$CC = \frac{\sum_{i \in \mathcal{G}, t \in \mathcal{T}} (f(i, t) - \bar{f})(f_{com}(i, t) - \bar{f}_{com}(i, t))}{\sqrt{\sum_{i \in \mathcal{G}, t \in \mathcal{T}} (f(i, t) - \bar{f})^2} \sqrt{\sum_{i \in \mathcal{G}, t \in \mathcal{T}} (f_{com}(i, t) - \bar{f}_{com}(i, t))^2}}. \quad (39)$$

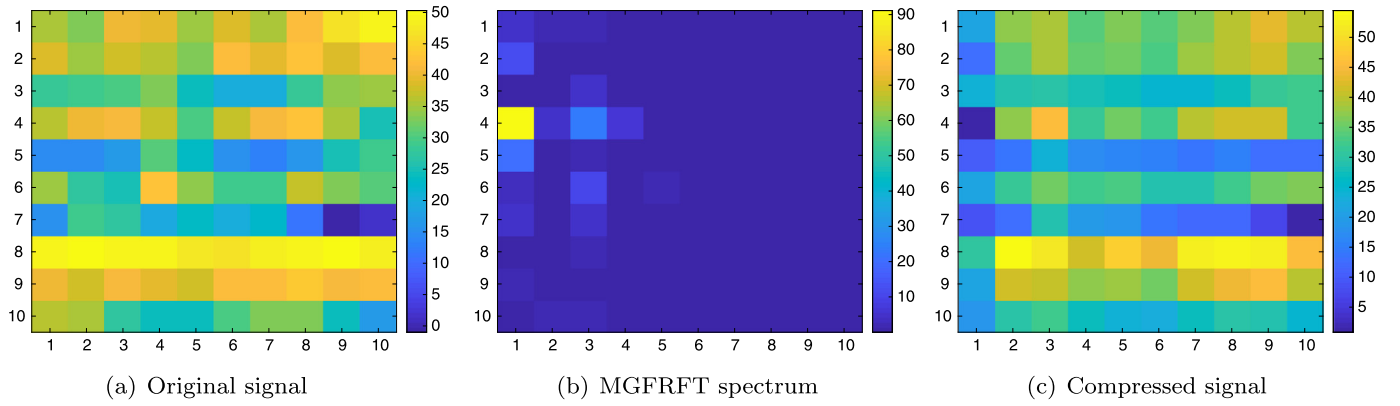


Fig. 5. An illustration of the compression process.

Table 1
RE, PSNR, NRMS, CC for different compression parameters.

$\alpha = 0.95$	$\gamma = 0.02$	$\gamma = 0.05$	$\gamma = 0.10$	$\gamma = 0.15$	$\gamma = 0.20$	$\gamma = 0.25$	$\gamma = 0.30$
RE(%)	10.49	8.56	7.00	5.97	5.18	4.52	3.96
PSNR(dB)	86.49	88.25	90.11	91.67	93.15	94.46	95.83
NRMS(%)	17.12	13.96	11.42	9.74	8.45	7.37	6.46
CC(%)	99.45	99.63	99.75	99.82	99.87	99.90	99.92
$\alpha = 0.85$	$\gamma = 0.02$	$\gamma = 0.05$	$\gamma = 0.10$	$\gamma = 0.15$	$\gamma = 0.20$	$\gamma = 0.25$	$\gamma = 0.30$
RE(%)	18.52	15.65	12.99	11.15	9.69	8.48	7.43
PSNR(dB)	89.10	86.80	85.11	85.59	86.93	88.20	89.60
NRMS(%)	30.21	25.53	21.19	18.19	15.82	13.84	12.12
CC(%)	98.27	98.77	99.15	99.38	99.53	99.64	99.72

RE and NRMS are proposed to measure the error between compressed signal and the original signal. For NRMS, in the denominator we subtract the mean to eliminate the effect of shifting. In both cases, smaller error implies better compression. PSNR and CC are proposed to measure the efficiency of compression. Larger values of PSNR and CC imply that the compressed signals essentially contain more information from the original signal.

We carry out various experiments with different compression ratio γ and transform fractional orders α . The result is shown in Table 1. The results show that even for extremely small γ , the compression only introduces a small relative error. Further, compared with the results shown in [43], the MGFRFT provides better transform basis and yields smaller error. At present we use enumeration method in the application to choose α and observe the influence of different values of α on the experimental results. For example, when we select $\gamma = 0.02$ in our application, if the goal of our application is to minimize the relative error, then α should be 0.95 according to Table 1. If we focus on PSNR, then α should be 0.85. The choice of α is through trial and error in most experiments. Tuning the fractional order α is also helpful to get more plausible compression.

6. Conclusions

In this paper, we present an approach to process multi-dimensional graph signal. We first propose the 2-D GFRFTs based on Laplacian matrix and adjacency matrix on graphs and extend them to multi-dimensional. Some properties such as index additivity and commutativity, are also given. Then to make implementations of our new transforms suitable for processing multi-dimensional signals, we compare the spectrum, directional characteristics and time cost obtained by one-dimensional transform and two-dimensional transform. The results show that the spectrum of our MGFRFT is well-defined, which avoids the double-valued spectrum of one-dimensional transforms that sometimes occurs. At the same time, the computation time is effectively reduced. Finally, we

apply the MGFRFT to data compression, using real-world data. The experiments reveal that compression based on our new transform is more efficient.

CRedit authorship contribution statement

Fang-Jia Yan: Conceptualization, Methodology, Validation, Writing – original draft. **Bing-Zhao Li:** Funding acquisition, Supervision, Writing – review & editing.

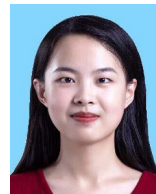
Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] K. Liao, Z. Yu, N. Xie, J. Jiang, Joint estimation of azimuth and distance for far-field multi targets based on graph signal processing, *Remote Sens.* 14 (5) (2022) 1110.
- [2] F. Gama, E. Isufi, G. Leus, A. Ribeiro, Graphs, convolutions, and neural networks: from graph filters to graph neural networks, *IEEE Signal Process. Mag.* 37 (6) (2020) 128–138.
- [3] M.W. Morency, G. Leus, Graphon filters: graph signal processing in the limit, *IEEE Trans. Signal Process.* 69 (2021) 1740–1754.
- [4] M.T. Schaub, Y. Zhu, J.-B. Seby, T.M. Roddenberry, S. Segarra, Signal processing on higher-order networks: livin' on the edge... and beyond, *Signal Process.* 187 (2021) 108149.
- [5] J. Miettinen, E. Nitzan, S.A. Vorobyov, E. Ollila, Graph signal processing meets blind source separation, *IEEE Trans. Signal Process.* 69 (2021) 2585–2599.
- [6] S.S. Saboksayr, G. Mateos, M. Cetin, EEG-based emotion classification using graph signal processing, in: 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2021, pp. 1065–1069.
- [7] E. Ceci, S. Barbarossa, Graph signal processing in the presence of topology uncertainties, *IEEE Trans. Signal Process.* 68 (2020) 1558–1573.
- [8] B. de Lathauwer, F. Navarro, B. Olivier, Localized fourier analysis for graph signal processing, *Appl. Comput. Harmon. Anal.* 57 (2022) 1–26.
- [9] W. Huang, T.A. Bolton, J.D. Medaglia, D.S. Bassett, A. Ribeiro, D. Van De Ville, A graph signal processing perspective on functional brain imaging, *Proc. IEEE* 106 (5) (2018) 868–885.

- [10] A. Sandryhaila, J.M.F. Moura, Big data analysis with signal processing on graphs: representation and processing of massive data sets with irregular structure, *IEEE Signal Process. Mag.* 31 (5) (2014) 80–90.
- [11] J. Domingos, J.M.F. Moura, Graph fourier transform: a stable approximation, *IEEE Trans. Signal Process.* 68 (2020) 4422–4437.
- [12] L. Yang, A. Qi, C. Huang, J. Huang, Graph Fourier transform based on ℓ_1 norm variation minimization, *Appl. Comput. Harmon. Anal.* 52 (2021) 348–365.
- [13] J. Jiang, C. Cheng, Q. Sun, Nonsubsampled graph filter banks: theory and distributed algorithms, *IEEE Trans. Signal Process.* 67 (15) (2019) 3938–3953.
- [14] R. Ramakrishna, H.-T. Wai, A. Scaglione, A user guide to low-pass graph signal processing and its applications: tools and applications, *IEEE Signal Process. Mag.* 37 (6) (2020) 74–85.
- [15] Z. Xiao, H. Fang, X. Wang, Distributed nonlinear polynomial graph filter and its output graph spectrum: filter analysis and design, *IEEE Trans. Signal Process.* 69 (2021) 1–15.
- [16] S. Chen, R. Varma, A. Sandryhaila, J. Kovačević, Discrete signal processing on graphs: sampling theory, *IEEE Trans. Signal Process.* 63 (24) (2015) 6510–6523.
- [17] S. Chen, A. Sandryhaila, J.M.F. Moura, J. Kovačević, Signal recovery on graphs: variation minimization, *IEEE Trans. Signal Process.* 63 (17) (2015) 4609–4624.
- [18] Y.H. Kim, Quantization-aware sampling set selection for bandlimited graph signals, *EURASIP J. Adv. Signal Process.* 2022 (1) (2022) 1–15.
- [19] N. Perraudin, B. Ricaud, D.I. Shuman, P. Vandergheynst, Global and local uncertainty principles for signals on graphs, *APSIPA Trans. Signal Inf. Process.* 7 (2018).
- [20] A. Agaskar, Y.M. Lu, A spectral graph uncertainty principle, *IEEE Trans. Inf. Theory* 59 (7) (2013) 4338–4356.
- [21] A. Mazarguil, L. Oudre, N. Vayatis, An uncertainty principle for lowband graph signals, *IEEE Signal Process. Lett.* 29 (2022) 727–731.
- [22] D.I. Shuman, S.K. Narang, P. Frossard, A. Ortega, P. Vandergheynst, The emerging field of signal processing on graphs: extending high-dimensional data analysis to networks and other irregular domains, *IEEE Signal Process. Mag.* 30 (3) (2013) 83–98.
- [23] D.I. Shuman, B. Ricaud, P. Vandergheynst, A windowed graph fourier transform, in: 2012 IEEE Statistical Signal Processing Workshop (SSP), IEEE, 2012, pp. 133–136.
- [24] D.K. Hammond, P. Vandergheynst, R. Gribonval, Wavelets on graphs via spectral graph theory, *Appl. Comput. Harmon. Anal.* 30 (2) (2011) 129–150.
- [25] A. Loukas, D. Foucard, Frequency analysis of time-varying graph signals, in: 2016 IEEE Global Conference on Signal and Information Processing (GlobalSIP), IEEE, 2016, pp. 346–350.
- [26] T. Kurokawa, T. Oki, H. Nagao, Multi-dimensional graph fourier transform, *arXiv preprint arXiv:1712.07811*, 2017.
- [27] Y. Wang, B. Li, Q. Cheng, The fractional fourier transform on graphs, in: 2017 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), IEEE, 2017, pp. 105–110.
- [28] Y. Wang, B. Li, The fractional Fourier transform on graphs: sampling and recovery, in: 2018 IEEE International Conference on Signal Processing (ICSP), IEEE, 2018, pp. 1103–1108.
- [29] J. Wu, F. Wu, Q. Yang, Y. Zhang, X. Liu, Y. Kong, L. Senhadji, H. Shu, Fractional spectral graph wavelets and their applications, *Math. Probl. Eng.* 2020 (2020).
- [30] F. Yan, W. Gao, B. Li, Windowed fractional fourier transform on graphs: fractional translation operator and hausdorff-young inequality, in: 2020 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), IEEE, 2020, pp. 255–259.
- [31] F. Yan, B. Li, Windowed fractional Fourier transform on graphs: properties and fast algorithm, *Digit. Signal Process.* 118 (2021) 103210.
- [32] B. Kartal, E. Özgünay, A. Koç, Joint time-vertex fractional fourier transform, *arXiv preprint arXiv:2203.07655*, 2022.
- [33] A. Natali, E. Isufi, G. Leus, Forecasting multi-dimensional processes over graphs, in: 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 5575–5579.
- [34] R. Varma, J. Kovačević, Sampling theory for graph signals on product graphs, in: 2018 IEEE Global Conference on Signal and Information Processing (GlobalSIP), IEEE, 2018, pp. 768–772.
- [35] C. Ozturk, H.M. Ozaktas, S. Gezici, A. Koç, Optimal fractional fourier filtering for graph signals, *IEEE Trans. Signal Process.* 69 (2021) 2902–2912.
- [36] Z. Ge, H. Guo, T. Wang, Z. Yang, Optimal fractional fourier filtering in time-vertex graphs signal processing, *arXiv preprint arXiv:2201.04335*, 2022.
- [37] A.I. Zayed, On the relationship between the Fourier and fractional Fourier transforms, *IEEE Signal Process. Lett.* 3 (12) (1996) 310–311.
- [38] R. Tao, Y. Li, Y. Wang, Short-time fractional fourier transform and its applications, *IEEE Trans. Signal Process.* 58 (5) (2009) 2568–2580.
- [39] M. Nassiri, G. Baghersalimi, Z. Ghassemlooy, Optical ofdm based on the fractional fourier transform for an indoor vlc system, *Appl. Opt.* 60 (9) (2021) 2664–2671.
- [40] H. Zhang, T. Shan, S. Liu, R. Tao, Optimized sparse fractional Fourier transform: principle and performance analysis, *Signal Process.* 174 (2020) 107646.
- [41] X. Liang, R. Xing, D. Shen, H. Wei, E. Liu, F. Ye, M. Lu, J. Wu, F. Zhang, R. Tao, Concise fractional Fourier transform based on a non-uniform order searching method for estimating physical parameters from Newton's rings, *Appl. Opt.* 61 (15) (2022) 4478–4485.
- [42] W. Imrich, S. Klavzar, D.F. Rall, *Topics in Graph Theory: Graphs and Their Cartesian Product*, CRC Press, 2008.
- [43] A. Sandryhaila, J.M.F. Moura, Discrete signal processing on graphs: graph Fourier transform, in: 2013 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2013, pp. 6167–6170.
- [44] G. Ortiz-Jiménez, M. Coutino, S.P. Chepuri, G. Leus, Sparse sampling for inverse problems with tensors, *IEEE Trans. Signal Process.* 67 (12) (2019) 3272–3286.
- [45] M.S. Kotzagiannidis, P.L. Dragotti, Sampling and reconstruction of sparse signals on circulant graphs—an introduction to graph-fri, *Appl. Comput. Harmon. Anal.* 47 (3) (2019) 539–565.
- [46] S.K. Kadambari, S.P. Chepuri, Learning product graphs from multidomain signals, in: 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 5665–5669.
- [47] G. Fracastoro, S.M. Fosson, E. Magli, Steerable discrete cosine transform, *IEEE Trans. Image Process.* 26 (1) (2016) 303–314.
- [48] A.P. Sanil, *Principles of Data Mining*, Kluwer Academic Publishers, 2003.



Fang-Jia Yan received the B.S. degree from the College of Science, Huazhong Agricultural University, Wuhan, China, in 2018. She is currently pursuing the Ph.D. degree with the School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, China. Her research interests include fractional Fourier transform, graph signal processing, and graph fractional Fourier transform.



Bing-Zhao Li was born in 1975. He received the B.S. degree from Shandong Normal University in 1998, and the M.S. and Ph.D. degrees from the Beijing Institute of Technology, in 2001 and 2007, respectively. He is currently a Professor with the School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, China. His current research interests include the theory and applications of fractional Fourier transform, linear canonical transform, and fractional

calculus.